Analysis of Kinetic Growth in Silicon Thermal Oxidation

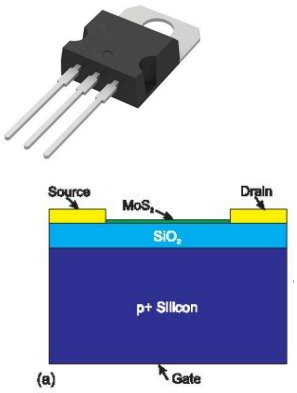
ETC 395

By:

Ethan Khan, Leo Bruno, Owen Lawrence

Creating ultra-thin insulating layers on silicon wafers is imperative for semiconductor devices, particularly in the realm of modern electronic devices where miniaturization and performance are paramount. This project delved into the growth of silicon dioxide (SiO_2) through a combination of experimental approaches and computational simulations. Our focus lies in investigating how factors such as temperature influence the thickness and properties of the SiO_2 layer.

Silicon dioxide (SiO_2), also known as silicon oxide, serves as a foundational element in the miniaturization and performance enhancement of modern electronic devices. The images to the right illustrate its significance in transistors, where it acts as a crucial insulating layer, facilitating efficient switching and signal amplification by separating conductive regions. Understanding the growth and properties of this essential insulating layer is pivotal for ongoing advancements in device miniaturization and performance.



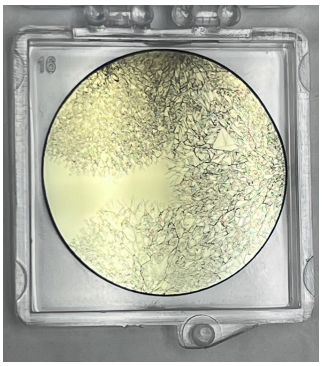
IRF520 MOSFET (Top). Showing Silicon Oxide layer in a transistor (bottom)

The thermal oxidation process, where a silicon wafer undergoes exposure to high temperatures in a controlled oxygen environment, enables the formation of a thin SiO_2 layer directly on the silicon surface. The growth rate and properties of this layer hinge significantly upon the oxidation temperature.

The Deal-Grove model, established in a seminal 1965 paper by Deal and Grove, provides a foundational framework for comprehending the relationship between oxidation temperature, time, and the resulting thickness of the SiO_2 layer. This model serves as a predictive tool for growth rates and activation energies, crucial parameters for optimizing the thermal oxidation process, and tailoring SiO_2 layer properties for specific device applications.

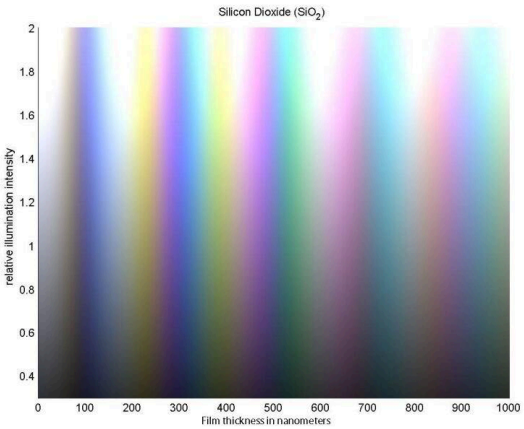
To start the experiment, the silicon wafers were given to us to perform oxidation. We are going to assume they underwent a cleaning and purification process beforehand. This is to ensure there are no contaminants present before oxidation. The setup for oxidation comprised loading the

cleaned silicon wafers onto a high-heat furnace. Before initiating the oxidation process, the furnace chamber was prepared with inert gas, such as nitrogen, to eliminate any residual oxygen and moisture. During the oxidation process, precise control over parameters was crucial. The ideal oxidation temperature, typically ranging between 800°C to 1200°C, was established depending on the targeted oxide thickness and growth rate. Oxygen or dry air was introduced into the furnace chamber at a controlled flow rate. The oxidation conditions were maintained for a predetermined duration, typically spanning from several minutes to several hours, contingent upon the desired oxide thickness.



Upon completion of the oxidation process, the furnace was gradually cooled to room temperature under an inert atmosphere. Subsequently, the oxidized wafers were cautiously removed from the furnace to prevent damage. Following this, the thickness of the grown oxide layer was measured using techniques such as **scanning electron microscopy (SEM)**. The quality of the oxide layer was evaluated through techniques like **atomic force microscopy (AFM)**.

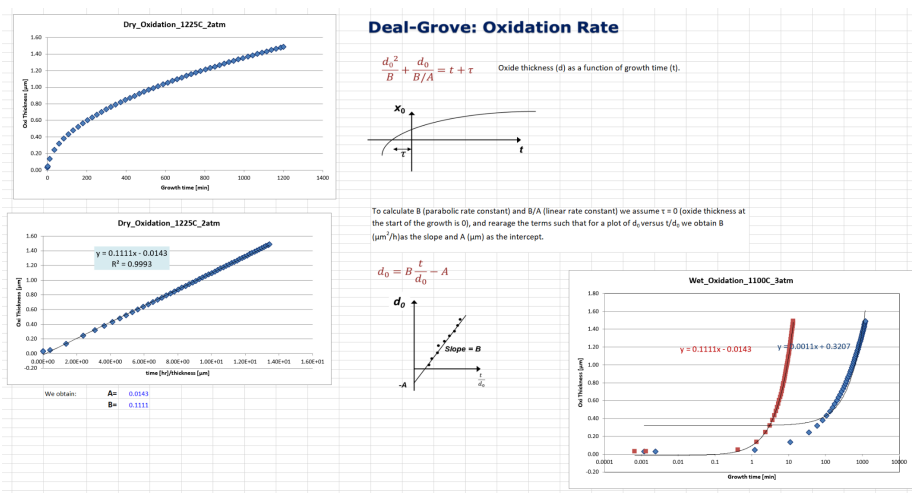
Throughout the experiment, several key parameters were monitored and controlled. These included the oxidation temperature, duration of the oxidation process, oxygen flow rate, as well as crystal orientation, and dopant concentration of the silicon wafers, if relevant. By adhering to this experimental procedure and meticulously controlling these parameters, silicon dioxide layers of precise thickness and quality could be reliably fabricated via thermal oxidation, thus contributing to the advancement of semiconductor device technology.



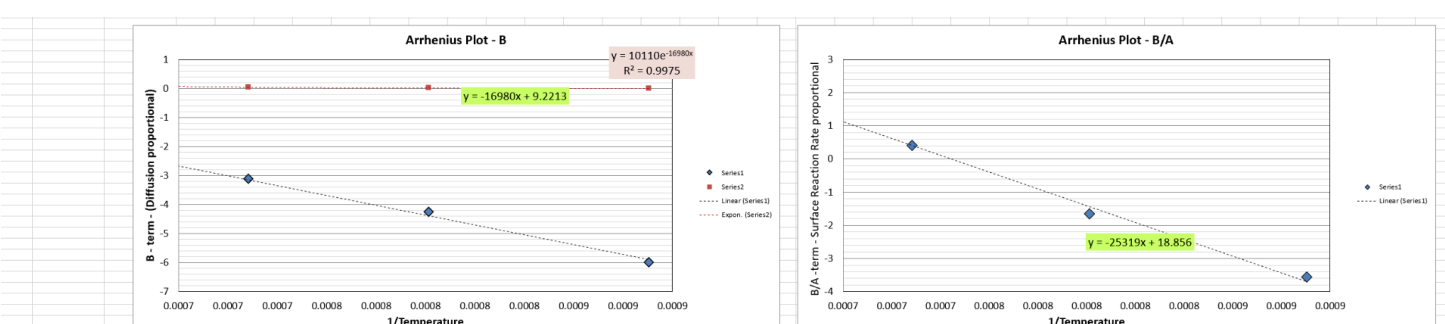
After the experiment, our results were satisfactory. For our experiment, the Silicon Wafer was in the High-Heat oven for 20 hours plus some extra time to reach room temperature. The silicon achieved oxidation at a constant temperature of 975℃. The oxide growth is indicated by the color in the

photograph above. The growth is shown with the crystal growth on the surface of the silicon wafer. The tint of the wafer shown is an indication of how much the wafer oxidized. Shown in the graph above our wafer had approximately 200-300 nanometers of growth.

To simulate silicon oxide growth, we utilized nanoHUB, a platform for accurate nanotechnology simulations. We input the parameters provided to us, including the oxidation process type (dry), growth time (20 hours), initial thickness of 30 nanometers, and various oxidation temperatures (T1: 850°C, T2: 975°C, T3: 1100°C, T4: 1225°C). Additionally, we specified atmospheric pressure (2 atm) and substrate orientation (111) to accurately model the thermal oxidation process. nanoHUB then generates oxide thickness versus time data, which are then graphed for each temperature, allowing us to observe how oxide thickness evolves over the duration of the oxidation process.



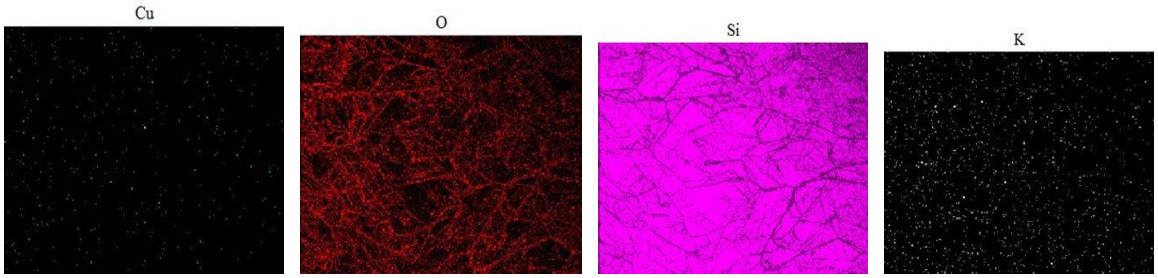
Moreover, nanoHUB facilitated the construction of Arrhenius plots, which illustrate the relationship between the rate of oxide growth and temperature. By plotting the natural logarithm of the oxide growth rate against the reciprocal of temperature (in Kelvin), we can apply the Arrhenius equation to extract activation energies associated with the thermal oxidation process.



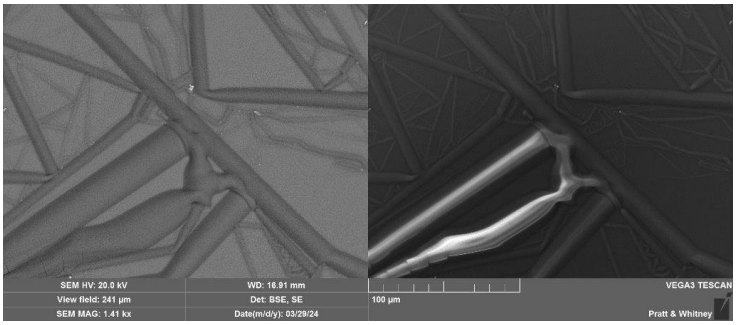
This approach provides valuable insights into the kinetics of oxide growth and allows for the characterization of reaction mechanisms underlying silicon oxidation.

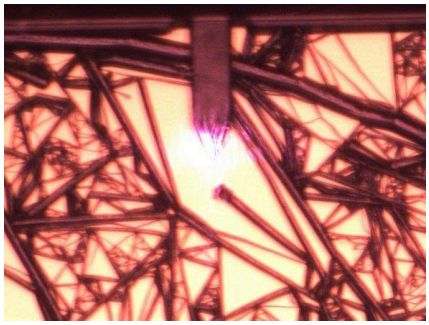
The methodology of employing Arrhenius plots to deduce activation energies is well-established in materials science, chemistry, and biology. J. Kohout's paper, "Modified Arrhenius Equation in Materials Science, Chemistry, and Biology," published in Molecules in 2021, discusses the application of the Arrhenius equation in diverse scientific disciplines. By leveraging this methodology, we can gain a deeper understanding of the thermal oxidation of silicon and elucidate the fundamental processes governing oxide growth, thereby facilitating advancements in semiconductor technology.

Following oxidation, the newly oxidized wafer underwent examination using a Scanning Electron Microscope (SEM) to investigate the surface features of the grown SiO_2 layer and identify any potential structures. SEM analysis provided information on the elemental composition of the SiO_2 film and the presence of nanowires, as depicted below.



Nanowires, tiny structures with diameters in the nanometer range, were observed on the surface of the oxide layer (Right). These nanowires hold potential for various electronic applications due to their unique electrical and physical properties.





In conclusion, thermal oxidation successfully formed a silicon dioxide (SiO_2) layer on the silicon wafer. SEM analysis revealed the presence of nanowire structures on the surface of the oxide

layer. The formation of a high-quality SiO_2 layer is crucial for the development of miniaturized and efficient electronic devices. Furthermore, the unexpected discovery of nanowires on the oxide layer opens up exciting possibilities for future exploration of their electrical and physical properties for potential electronic applications.

The simulated data obtained from the Deal-Grove model provided valuable insights into the relationship between oxidation temperature and the growth rate of the SiO_2 layer. This information can be used to refine future oxidation processes for precise control over the oxide layer thickness. For future optimization, the data collected from the oxidation models and Arrhenius plots offer valuable insight into optimizing the thermal oxidation process, leading to better efficiency in the Semiconductor Fabrication Process.

References Cited:

- B. E. Deal, A. S. Grove. “General Relationship for Thermal Oxide Growth”. Fairchild Semiconductor, Journal of Applied Physics. Palo Alto, California. (1965). 36-12.
- Cao, Shuqing (Victor), et al. “Process Lab:Oxidation.” *nanoHUB.Org*, 8 Oct. 2006, nanoHUB.org/resources/prolabox.
- Arrhenius Equation. (2020, September 10). Retrieved April 27, 2021, from https://chem.libretexts.org/@go/page/1443
- J. Kohout. “Modified Arrhenius Equation in Material Science,Chemistry and Biology”. Molecules. (2021).
- Reliability of Single-Layer MoS2 Field-Effect Transistors with SiO2 and hBN Gate Insulators. (n.d.-b).
- Effect of native oxide on stress in silicon nanowires. (n.d.-a). https://pubs.acs.org/doi/10.1021/acsanm.2c02983